

Theorem: π^2 (and therefore π) is irrational (As if π were rational π^2 would be too). Because rational numbers are exactly those whose decimals repeat (proven in Level 2), π 's decimals never repeat.

Levels recommended for proof: 3-5

Proof:

Start by assuming $\pi^2 = \frac{a}{b}$, we aim to derive a contradiction. Define the sequence of integrals I_n as follows:

- $I_0 = 2$
- $I_1 = 4b$
- $I_n = \frac{b^n}{n!} \int_0^\pi x^n (\pi - x)^n \sin(x) dx \quad (n \geq 2)$

We can bound I_n as follows: It is non-negative since everything in the integral is non-negative and so is the coefficient, and $x^n(\pi - x)^n$ is bounded above by $\pi^n \pi^n = \pi^{2n}$ whenever x is between 0 and π , and therefore the integral can be bounded above by $\int_0^\pi \pi^{2n} \sin(x) dx = \pi^{2n} \int_0^\pi \sin(x) dx = 2\pi^{2n}$. Therefore we can bound I_n by $\frac{2(\pi^2 b)^n}{n!}$. Note that when, for example, $n > 2\pi^2 b$, the terms $\frac{2(\pi^2 b)^n}{n!}$ will be at most half of the previous term (since the numerator will be multiplied by $\pi^2 b$ and the denominator by more than twice that), therefore these terms eventually tend to 0. Since I_n is bounded above by these terms and never negative, I_n also tends to 0.

The idea of the proof will be to show that if $\pi^2 = \frac{a}{b}$ then I_n is an integer for all n , which is a problem since I_n is always strictly between 0 and something that tends to 0 (and therefore is less than 1), so I_n can never be an integer and this will be our contradiction.

Now we will do a classic A level further maths problem where we show from algebra that for n at least 2, $I_n = 2b(2n - 1)I_{n-1} - abI_{n-2}$, which would mean that I_n is always an integer (since a , b , n , and I_0, I_1 are all integers) so we will be done.

It turns out that $I_n = \frac{b^n}{n!} \int_0^\pi x^n (\pi - x)^n \sin(x) dx$ for $n=0$ and $n=1$. The proof is that for $n=0$, the integral simplifies to $\frac{b^0}{0!} \int_0^\pi x^0 (\pi - x)^0 \sin(x) dx = \int_0^\pi \sin(x) dx = 2$ (it's kind of cool that the area under one of the waves of the sine function is exactly 2 isn't it), and for $n=1$, the integral simplifies to $b \int_0^\pi x(\pi - x) \sin(x) dx = b\pi \int_0^\pi x \sin(x) dx - b \int_0^\pi x^2 \sin(x) dx$. Integration by parts on the first integral setting $u = x, \frac{dv}{dx} = \sin(x)$ so $\frac{du}{dx} = 1, v = -\cos(x)$ gives that it is exactly equal to $-[x\cos(x)]_0^\pi - \int_0^\pi -\cos(x) dx = -(-\pi - 0) - 0 = \pi$. Also, $\int_0^\pi x^2 \sin(x) dx$ can be integrated by parts if we set $u = x^2, dv = \sin(x) dx$ so $du = 2x dx, v = -\cos(x)$, meaning that the integral is then equal to $-[x^2 \cos(x)]_0^\pi - \int_0^\pi -2x \cos(x) dx = -(-\pi^2 - 0) + \int_0^\pi 2x \cos(x) dx = \pi^2 + \int_0^\pi 2x \cos(x) dx$. Integrating the second integral by parts, setting $u = 2x, \frac{dv}{dx} = \cos(x)$ so $\frac{du}{dx} = 2, v = \sin(x)$, we get that $\int_0^\pi 2x \cos(x) dx = [2x \sin(x)]_0^\pi - \int_0^\pi 2 \sin(x) dx = (0 - 0) - 4 = -4$, therefore the original integral $\int_0^\pi x^2 \sin(x) dx$ is equal to $\pi^2 - 4$. Now $I_1 = b\pi(\pi) - b(\pi^2 - 4) = 4b$, as required.

Now the final step is to prove the recurrence relation $I_n = 2b(2n - 1)I_{n-1} - abI_{n-2}$ for n at least 2.

Consider $\int_0^\pi x^n(\pi - x)^n \sin(x) dx$. We will do this by parts. Set $u = x^n(\pi - x)^n$, $\frac{dv}{dx} = \sin(x)$. By the product and chain rules, $\frac{du}{dx} = nx^{n-1}(\pi - x)^n - nx^n(\pi - x)^{n-1} = x^{n-1}(\pi - x)^{n-1}[n(\pi - x) - nx] = nx^{n-1}(\pi - x)^{n-1}(\pi - 2x)$. By the integration by parts formula $\int_0^\pi u \frac{dv}{dx} dx = [uv]_0^\pi - \int_0^\pi v \frac{du}{dx} dx$, but u vanishes (is 0) at 0 and π and therefore uv does too. Therefore $\int_0^\pi u \frac{dv}{dx} dx = - \int_0^\pi v \frac{du}{dx} dx$. Therefore, since $v = -\cos(x)$, $\int_0^\pi x^n(\pi - x)^n \sin(x) dx = n \int_0^\pi x^{n-1}(\pi - x)^{n-1}(\pi - 2x) \cos(x) dx$. Now we want to integrate this by parts again so work out the derivative of $x^{n-1}(\pi - x)^{n-1}(\pi - 2x)$. By the product rule on the last term, this is equal to $-2x^{n-1}(\pi - x)^{n-1} + (\pi - 2x) \frac{d}{dx}[x^{n-1}(\pi - x)^{n-1}]$, but this last derivative is just the $n-1$ version of something whose derivative we already know, so $\frac{d}{dx}[x^{n-1}(\pi - x)^{n-1}] = (n-1)x^{n-2}(\pi - x)^{n-2}(\pi - 2x)$. Therefore the original derivative we wanted, ie the derivative of $x^{n-1}(\pi - x)^{n-1}(\pi - 2x)$, is $-2x^{n-1}(\pi - x)^{n-1} + (\pi - 2x)^2(n-1)x^{n-2}(\pi - x)^{n-2}$. Therefore for $\int_0^\pi x^{n-1}(\pi - x)^{n-1}(\pi - 2x) \cos(x) dx$ we set $u = x^{n-1}(\pi - x)^{n-1}$, $\frac{dv}{dx} = \cos(x)$, so now we have $v = \sin(x)$, $\frac{du}{dx} = -2x^{n-1}(\pi - x)^{n-1} + (\pi - 2x)^2(n-1)x^{n-2}(\pi - x)^{n-2}$. This time v vanishes at 0 and π so we reduce to $-\int_0^\pi v \frac{du}{dx} dx$

$$= \int_0^\pi [2x^{n-1}(\pi - x)^{n-1} - (\pi - 2x)^2(n-1)x^{n-2}(\pi - x)^{n-2}] \sin(x) dx. \text{ By some algebra, this is } \int_0^\pi [2x^{n-1}(\pi - x)^{n-1} - (\pi^2 - 4x(\pi - x))(n-1)x^{n-2}(\pi - x)^{n-2}] \sin(x) dx. \text{ This is therefore equal to } \int_0^\pi [(2 + 4(n-1))x^{n-1}(\pi - x)^{n-1} - (\pi^2)(n-1)x^{n-2}(\pi - x)^{n-2}] \sin(x) dx.$$

Ok so putting all this together

$$\begin{aligned} I_n &= \frac{b^n}{n!} \int_0^\pi x^n(\pi - x)^n \sin(x) dx = \frac{b^n}{(n)!} n \int_0^\pi x^{n-1}(\pi - x)^{n-1}(\pi - 2x) \cos(x) dx \\ &= \frac{b^n}{(n-1)!} \left[\int_0^\pi [(2 + 4(n-1))x^{n-1}(\pi - x)^{n-1} - (\pi^2)(n-1)x^{n-2}(\pi - x)^{n-2}] \sin(x) dx \right] \\ &= b \frac{b^{n-1}}{(n-1)!} \int_0^\pi [(2 + 4(n-1))x^{n-1}(\pi - x)^{n-1}] \sin(x) dx - \frac{b^2 \pi^2}{n-1} \frac{b^{n-2}}{(n-2)!} \left[\int_0^\pi [(n-1)x^{n-2}(\pi - x)^{n-2}] \sin(x) dx \right] \\ &= b I_{n-1} (2 + 4(n-1)) - b^2 \pi^2 \frac{b^{n-2}}{(n-2)!} \left[\int_0^\pi [x^{n-2}(\pi - x)^{n-2}] \sin(x) dx \right] \\ &= 2b(2n-1)I_{n-1} - ab I_{n-2} \text{ where we have used the fact that } \pi^2 = \frac{a}{b}. \text{ This completes the proof of the formula so we are done.} \end{aligned}$$